what we can muster with our current understanding of string theory's equations.

As a matter of fact, I suspect, as do many other string theorists, that the tiny masses in Tables 12.1 and 12.2 arise in string theory much as they do in the standard model. Recall from Chapter 9 that in the standard model, a Higgs field takes on a nonzero value throughout all space, and the mass of a particle depends on how much drag force it experiences as it wades through the Higgs ocean. A similar scenario likely plays out in string theory. If a huge collection of strings all vibrate in just the right coordinated way throughout all of space, they can provide a uniform background that for all intents and purposes would be indistinguishable from a Higgs ocean. String vibrations that initially yielded zero mass would then acquire tiny nonzero masses through the drag force they experience as they move and vibrate through the string theory version of the Higgs ocean.

Notice, though, that in the standard model, the drag force experienced by a given particle—and hence the mass it acquires—is determined by experimental measurement and specified as an input to the theory. In the string theory version, the drag force—and hence the masses of the vibrational patterns—would be traced back to interactions between strings (since the Higgs ocean would be made of strings) and should be *calculable*. String theory, at least in principle, allows all particle properties to be determined by the theory itself.

No one has accomplished this, but as emphasized, string theory is still very much a work in progress. In time, researchers hope to realize fully the vast potential of this approach to unification. The motivation is strong because the potential payoff is big. With hard work and substantial luck, string theory may one day explain the fundamental particle properties and, in turn, explain why the universe is the way it is.

The Fabric of the Cosmos According to String Theory

Even though much about string theory still lies beyond the bounds of our comprehension, it has already exposed dramatic new vistas. Most strikingly, in mending the rift between general relativity and quantum